Mikhail Razakov

Compiler and Runtime Engineer

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I build UniversalToolchain/Wist2, a modular .NET compiler/runtime platform with multiple IRs, a reference interpreter, a CIL/DynamicMethod backend and a verifier-gated AIR → SSA → AIR route. I also work on conservative program analysis, C++23 components and x86 code generation.

AIR → SSA → AIR
Optimizations and structural verifiers

1st of 49
PS-form Analyzer, 104/104

1,358 / 1,358 tests
IR transforms, interpreter/CIL parity and contracts

EXPERIENCE

2024 - PRESENT

UniversalToolchain / Wist2 - creator and primary developer

- Designed Source -> AST -> Bytecode -> AIR -> execution, a reference interpreter and a CIL/DynamicMethod backend.
- Implemented opt-in AIR → SSA → AIR with constant folding, SCCP-lite, branch folding, unreachable cleanup and DCE.
- Added capability contracts, structural verifiers and interpreter/CIL parity checks for supported semantics.

JUL - AUG 2026

MCST - compiler engineering intern, LLVM track

- Developed a shared graph core and C++23 analysis components for the LLVM track.
- Implemented RPO with cycle detection, Dijkstra, Tarjan SCC and Dinic; used CMake, warnings-as-errors, ASan/UBSan, clang-tidy, explicit includes and reproducible I/O tests.

SELECTED PROJECTS

PS-form Memory Dependence Analyzer

C17 / PROGRAM ANALYSIS

Conservative inter-iteration memory-dependence analysis. Ranked 1st of 49 with the only 5.0/5.0 and 104/104 result; exact oracle plus randomized and metamorphic tests.

NASM IA-32 and x86-64 codegen

NASM COURSE

NASM IA-32: cdecl, stack frames, libc, structures/addressing and x87. Separate x86-64 SysV emitter lab: liveness, live intervals, linear scan, iced-x86 and interpreter-vs-native validation.

CORE SKILLS

Compiler / IR
parsing, AST, Bytecode, AIR, CFG, SSA, CIL/DynamicMethod

Analysis / optimization
dominance, liveness, SCCP-lite, DCE, memory-dependence analysis

Correctness
structural verifiers, exact oracles, differential and metamorphic testing

Low-level
NASM IA-32, cdecl, stack frames and x87; x86-64 SysV codegen, objdump/disassembly

Languages / tools
C#/.NET, C++23, C17, Python; CMake, GitHub Actions, sanitizers, clang-tidy

EDUCATION AND AVAILABILITY

Software Engineering student at HSE University. Moscow / remote. From September 2026: up to 20 hours per week.

TECHNICAL COMMUNICATION

Taught about 50 students; authored a practical NASM IA-32 course, documentation and compiler/runtime technical materials.

RECOGNITION

HSE Olympiad prize-winner in competitive and industrial programming; top MEPhI Junior score in Engineering Sciences (2025) and the IT section (2026); Baltic competition Grand Prize.